



Reactive Table

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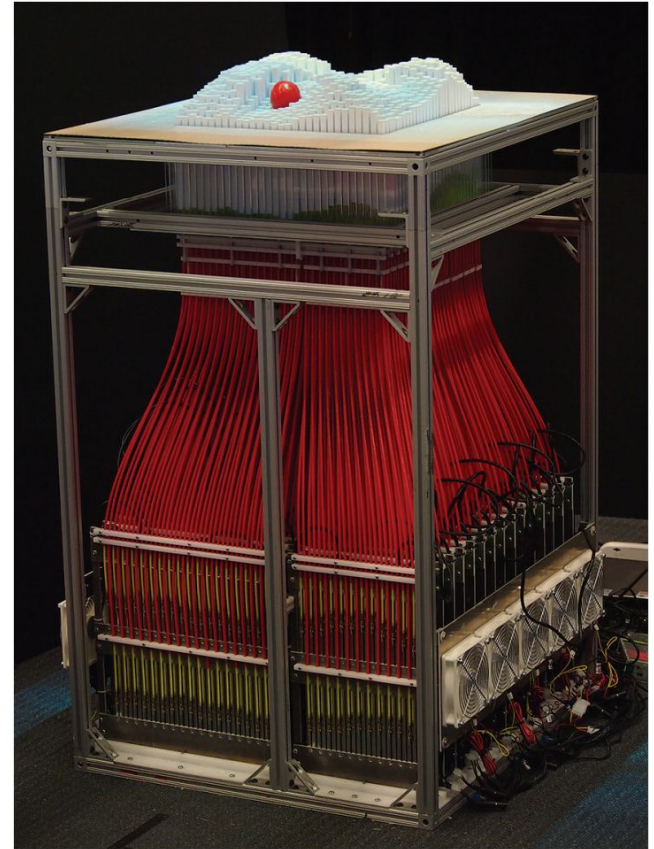
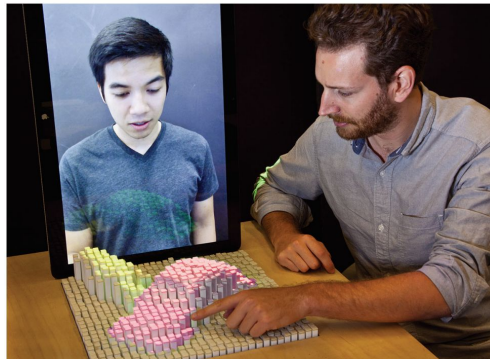
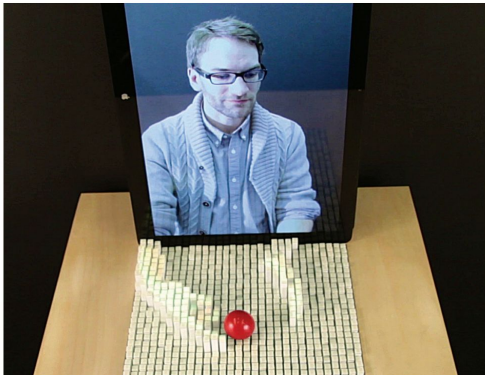
Project Description



Our project focuses on designing and building a form-changing table that can physically reshape its surface in real time. The tabletop is constructed from an array of small, independently actuated square modules that move up and down. By precisely controlling these modules, the table can dynamically create 3D surfaces, replicate objects placed on it, or adapt to different user needs.

Precedent Work

Researchers at MIT have created a similar shape shifting table which uses an array of motorized pins to manipulate the table's surface geometry. The current design combines shape rendering with visual feedback and depth sensing to support interaction techniques and mimic shapes. This allows people to remotely interact with objects and other people.



Precedent Work

Research in the Gonsher Lab investigates the role of robotic furniture as artificial chatbot assistants are becoming increasingly common. The goal is for this furniture to be integrated into the environment, allow users to have telepresence in other spaces, and have objects that can be responsive to a variety of use cases. Integration of interactive displays will allow for greater tactile and visual interaction with users.



Specifications/Standards



Institute of Electrical and Electronics Engineers

- [1872-2015: IEEE Standard Ontologies for Robotics and Automation](#)
- [IEEE 1451 and 1588 - Transducer Functionality and Precision Time Protocol](#)

American Society for Testing and Materials

- [ISO/ASTM 52910 - Guidelines for Design for AM \(ISO/ASTM 52910\)](#)

Society of Automotive Engineers

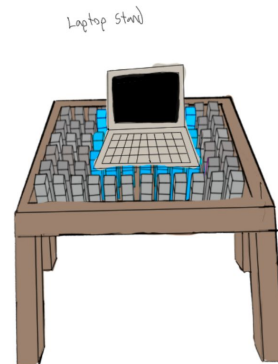
- [ARP5812 - Actuators, Linear Mechanical, General Specification Guide](#)

Use Cases

- **Vacuum Forming** - Adaptable, vacuum-sealable form creation using various actuator arrays
- **Interactive Electronics** - Keyboard/button functionality integrated with personal electronic devices

Other ideas:

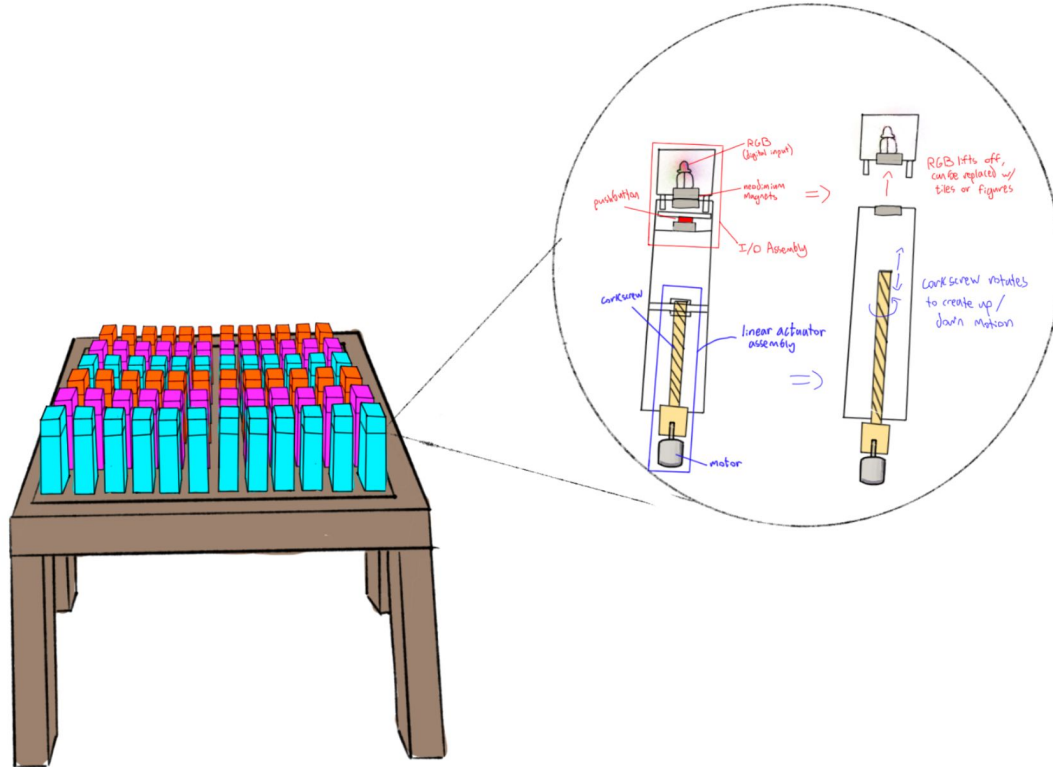
- **Video/Board Games** - interactive gaming experiences that allow for tactile feedback and input
- **Remote Interactions** - Integration with facetime to allow remote control/connection of separated devices
- **Interactive Art Display** - Table can make sinusoidal or pleasing patterns when idle
- **Household Uses** - Cubbies/containers, pencil holders, computer stands, TV/stereo system remote



Subsystems / Implementation Plan

- **Control**
 - Raspberry Pi → Arduino Uno → Arduino Nano -> L298 Motor Driver -> 2 motors
- **Mechanical Movement**
 - 3x3 (9 unit square) array of motors attached to linear actuators to allow for moving columns
- **Input/Output**
 - LED lighting for each column
 - Buttons on each column to allow for responsive programs
 - Possibly a small display/phone input to change between different modes

Subsystems Diagram and Demonstration



Provisional Budget

**Note: A portion of these items are either already on hand, meaning they do not contribute to the net total, or are non-critical.*

Item	Link to purchase	Items per Unit	Price (each)	Number of Units	Total	Obtainable From School?	Order Placed	Received
M4 100mm screws	https://www.mcmaster.com/	25	10.56	1	10.56	☐	☒	☐
M4 locknut	https://www.mcmaster.com/	100	14.05	1	14.05	☐	☒	☐
M4 square nut	https://www.mcmaster.com/	100	8.8	1	8.8	☐	☒	☐
Threadlocker (lock-tite)	https://www.amazon.com/	50ml	9.99	1	9.99	☒	☐	☐
B94ZZ ball bearing	https://www.mcmaster.com/	1	9.2	1	9.2	☐	☒	☐
M3 10mm screws	https://www.mcmaster.com/	100	12.19	1	12.19	☐	☒	☐
Raspberry Pi	https://a.co/d/5G8W8	1	59.99	1	59.99	☒	☒	☐
Raspberry Pi Power Supply	https://www.amazon.com/	1	9.99	1	9.99	☐	☒	☐
Raspberry Pi Micro USB to USB-A adapter	https://www.amazon.com/	1 adapter	14.39	1	14.39	☐	☒	☐
Motor Encoder					0	☐	☐	☐
Motors	https://www.amazon.com/	12	10.99	2	21.98	☐	☒	☒
Drivers	https://www.amazon.com/	4	9.98	2	19.96	☐	☒	☐
ABS 3D Printing Filament			25.99	1	25.99	☒	☐	☐
Soldering pin extenders	https://www.amazon.com/		4.99	1	4.99	☒	☐	☐
ATTiny Controller	https://www.digikid.com/		1.66	16	26.56	☐	☐	☐
5050 LED Strip	https://www.ledlight.com/		59.50	1	59.5	☒	☐	☒
Display	https://www.buydisplay.com/		3.25	1	3.25	☒	☐	☐
Arduino Nano Pack	https://store-usa.arduino.cc/		36.3	1	36.3	☐	☐	☐
LED Light Strip	https://www.amazon.com/		23.79			☐	☐	☐
NFC Cards	https://www.amazon.com/	20	6.99			☐	☐	☐
NFC Readers	https://www.amazon.com/	10	32	2		☐	☐	☐
Micro Motor	https://www.digikid.com/	Unsure	5.88	1	12.26	☐	☒	☐
USB Splitter	https://www.amazon.com/					☐	☐	☐
DC Buck Converter	https://www.amazon.com/	5	7.99	1	7.99	☐	☐	☐
Overall Total					303.15			
Net Total (w/ parts on hand)					196.24			

Provisional Timeline



Week of 10/26: Present CDR with CAD models of linear actuator, iterate on linear actuator sizing and parts, finish the coding for one linear actuator.

Week of 11/2: Mechanically design and electronically code a sensor/feedback system for the linear actuator to react to applied pressure.

Week of 11/9: Iterate on sensor/feedback loop, and begin to write final report.

Week of 11/16: Continue writing the final report, design/make a table/frame to fit the linear actuators in and figure out how to assemble them all next to each other.

Week of 11/23: Lowkey nothing because Thanksgiving break - make any finishing touches needed, and wrap up the final report.

Week of 11/30: Present our findings! Add finishing touches/recommended edits to the final report and submit.

Design Criteria and Evaluation



Design Criteria:

- **Functionality:** The tabletop must generate 3D surfaces with enough resolution, range, and strength to support everyday use.
- **Mechanical Design:** Actuators must be compact, modular, and durable for reliable long-term operation.
- **Electronics:** The system should coordinate many actuators efficiently with synchronized, lag-free motion.
- **Aesthetic:** The surface should function as a normal table when flat and move quietly and smoothly when active.

Evaluation:

- Test the ability of the table to model several 3D surfaces.
- Test linear actuator repeatability by changing table top surface several times.
- Whatever we define as “Proper functional behavior” (responsiveness, ease of use, no bugs, expected behavior, ect.)

Publication Venues



- ICRA, Atlanta 2025: IEEE International Conference on Robotics and Automation
- Robotics Science and Systems; Sydney, Australia 2026
- Journal of Field Robotics